Aga Khan University Examination Board

Notes from E-Marking Centre on HSSC-II Physics Examination April/ May 2019

Introduction

This document has been produced for the teachers and candidates of Higher Secondary School Certificate (HSSC) Part II Physics. It contains comments on candidates' responses to the 2019 HSSC-II Examination, indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses which support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

General Observations

This year candidates performed well on questions related to the electric field and electric intensity, power dissipation, force on a moving charged particles, electrical instruments, law of electromagnetic induction, AC and DC generators, three phase AC supply, mechanical properties of solids, semi-conductor devices, detection of ionising radiation, uses of radiation for medical diagnosis and therapy, Gauss's law, word problem based on transformer, quantum theory and photoelectric effect. Whereas low-scoring candidates did not perform well in questions based on electromagnetic waves, special theory of relativity, excitation and ionisation potential, energy stored in a capacitor, energy stored in an inductor, as well as, had problem in articulating their answers to differences, reasoning and discussion type questions.

Detailed Comments:

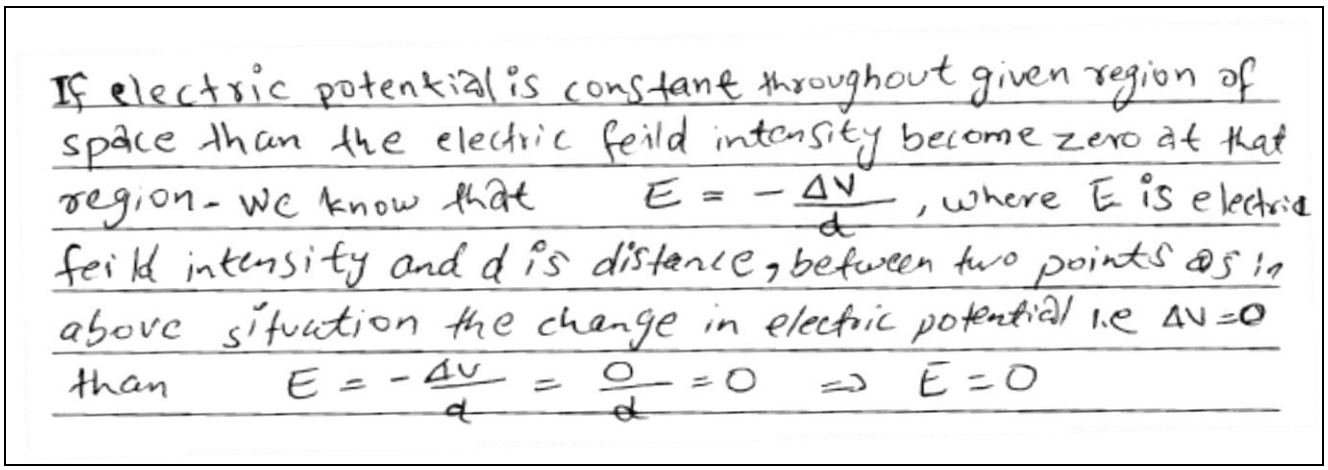
Constructed Response Questions (CRQs)

Question 1:

If the electric potential is constant throughout a given region of space, then is it possible that the electrical field intensity becomes zero in this region? Justify your answer with the help of a mathematical evidence.

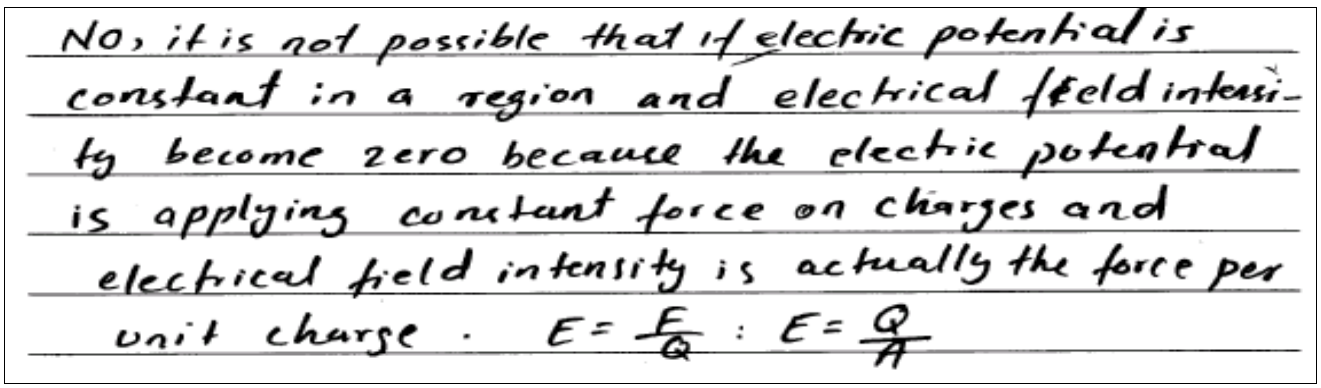
Better responses presented correct justification that when electric potential is constant throughout a given region of space, then the electrical field intensity becomes zero. High-scoring candidates wrote in terms of mathematical formula as $E = - \Delta V/\Delta r$. Some of the candidates wrote that the rate of change of potential with distance and negative sign shows the direction of 'E' along the direction of decreasing potential. Few candidates wrote that potential different 'V' is constant throughout the region, so $\Delta V = 0$, hence, $E = 0$.

Example:



Weaker responses presented incorrect justification. They wrote that electric potential is not constant throughout a given region of space; therefore, it is not possible that the electrical field intensity becomes zero in the region. Few of the low-scoring candidates wrote their answer in negation and rarely mentioned one or two correct points with correct meaning.

Example:



Question 2:

The given table shows the information about two filament lamps that are connected in an electrical circuit.

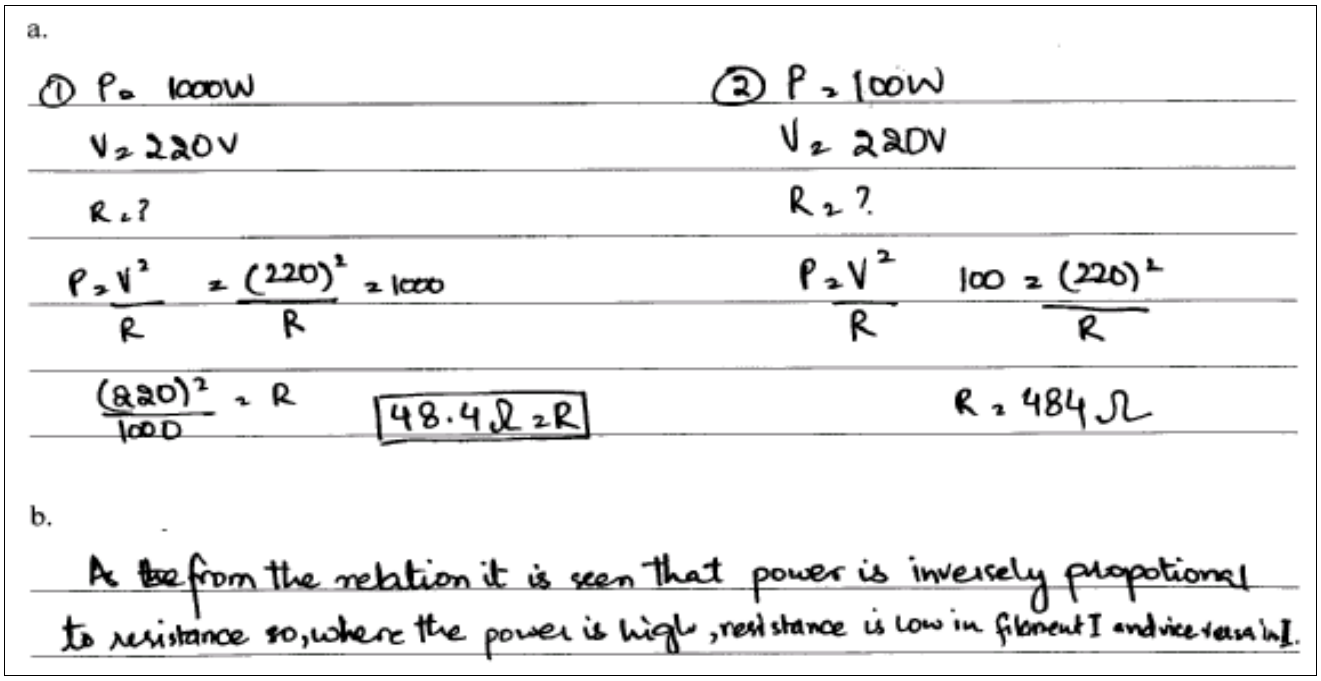
Type of Filament Lamp	Power Consumption	Operating Voltage
Filament lamp I	1000 W	220 V
Filament lamp II	100 W	220

a. Calculate the resistance of both the filament lamps.

b. Why the resistance of filament lamp I is different from filament lamp II?

Better responses calculated the resistance of both the given filament lamps and explained that the resistance of both the filaments is different. In part (a), high-scoring candidates used the correct formula of power dissipation, $P = \frac{V^2}{R}$, substituted the correct values and got the correct answers for both the filaments, i.e. R = 48.4 Ω and R = 484 Ω respectively. In part (b), candidates wrote that resistance of the filament lamp-I is inversely proportional to power or resistivity of filament lamp-II is more because at low resistance more power is consumed. Few candidates wrote that power is not equal and greater power is consumed in-I and both have different power consumption.

Example:



Weaker responses failed to calculate the resistance of both the given filament lamps and also were unable to explain that the resistance of both the filaments is different. In part (a), most of the low-scoring candidates wrote the incorrect formula of power dissipation, therefore, they were unable to calculate the correct value of resistance. In part (b), candidates also failed to reason out that resistance of filament lamp-I is different from filament lamp-II and hence, wrote the reference of semi-conductor and Ohm's law.

Example:

a.

Data:

Voltage$_1$ = 220V$_1$

Voltage$_2$ = 220V$_2$

Current$_1$ = 1000W$_1$

Current$_2$ = 100W$_2$

R_1 = ?

R_2 = ?

Working formula = $R = \frac{V}{I}$

$R_1 = \frac{V_1}{I_1}$

$R_1 = \frac{220V}{1000W}$

$R_1 = 0.22\Omega$

$R_2 = \frac{V_2}{I_2}$

$R_2 = \frac{220V}{100W}$

$R_2 = 2.2\Omega$

b.

Because filament lamp is a semi-conductor. which not obays ohms. Law.

Question 3:

Three particles are projected into a magnetic field. The velocities of these particles are at right angle to the magnetic field.

Identify the charge on these particles, if

a. two of these particles deflect in the opposite direction.
b. the third particle does not deflect in any direction.

Better responses correctly identified the charge on the particles, when two of the particles deflected in the opposite direction and the third particle does not deflect in any direction. In part (a), high-scoring candidates wrote that one particle is positive while the other is negative. In part (b), candidates mentioned that the third particle has no charge/ neutral.

Example:

a.

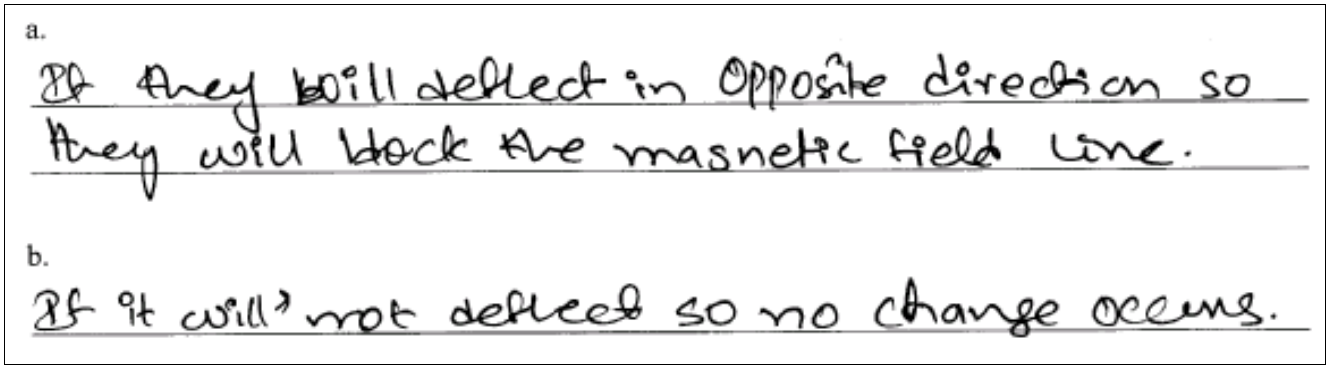
The two particles are charged particles of opposite nature one is positive another negative

b.

The third particle is electrically neutral it don't have any charge.

Weaker responses mentioned the irrelevant terms in their descriptions, i.e. magnetic field line, electric charges and charged particles. Few of the low-scoring candidates wrote the formula of kinetic energy and compared it with the electric field and potential difference of the electron.

Example:



Question 4:

An ammetre is connected with an electrical circuit in series.

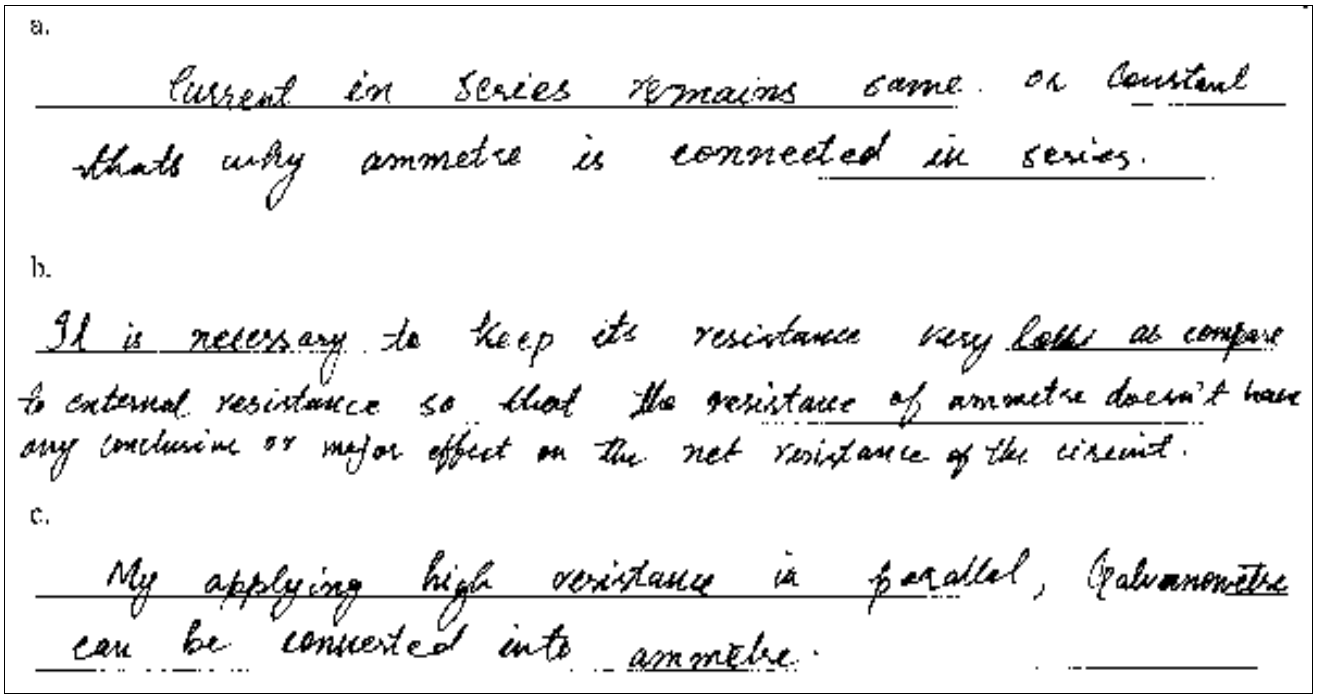
a. Why is it connected in series?
b. Why is it necessary to keep its resistance very low?
c. How can a galvanometer be converted into an ammetre?

Better responses wrote the correct reasons in all three parts. In part (a), high-scoring candidates wrote that all the current to be measured must pass through the ammetre, so it is connected in series because current in the series remains constant.

In part (b), candidates wrote that a low resistance in the circuit causes the current to decrease. Hence, maximum current can pass through the shunt resistance.

In part (c), candidates wrote that when a low resistance, called the shunt resistance, is to be connected parallel to the coil of a galvanometer, it gets to converted it into an ammeter.

Example:


a.

Current in series remains same or constant that why ammetre is connected in series.

b.

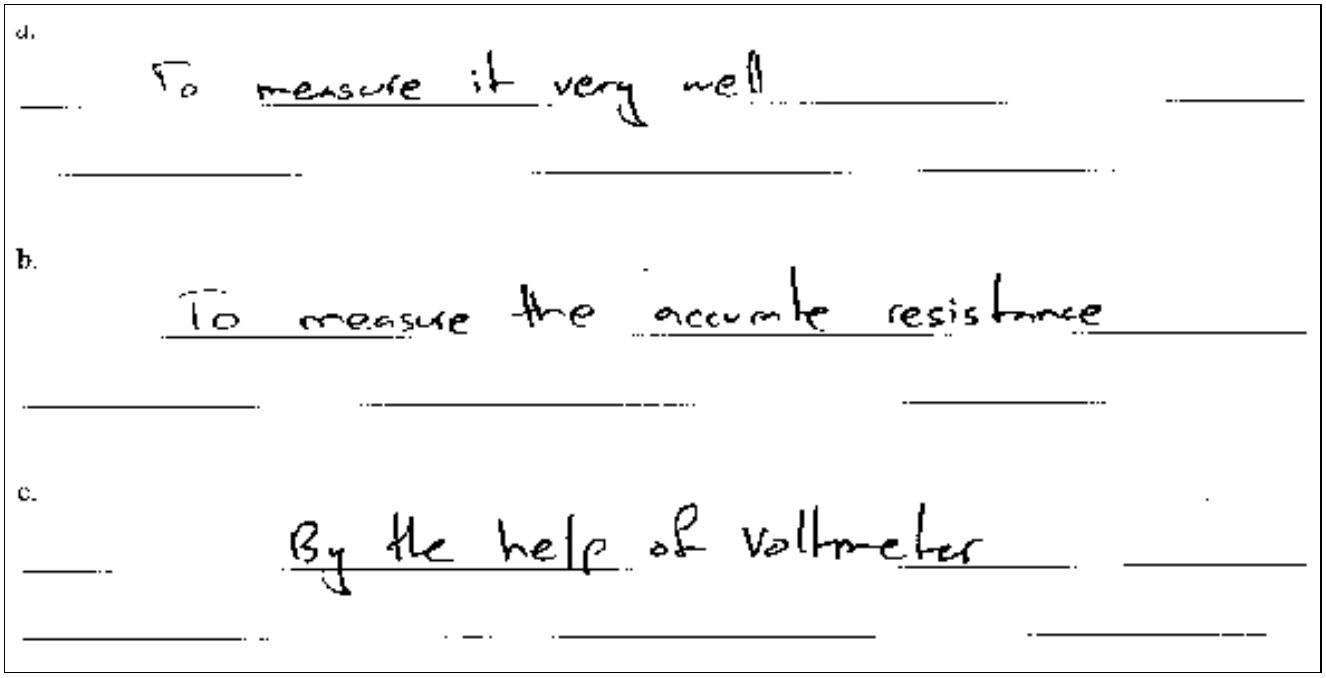
It is necessary to keep its resistance very low as compare to external resistance so that the resistance of ammetre doesn't have any conclusive or major effect on the net resistance of the circuit.

c.

My applying high resistance in parallel, Galvanometre can be converted into ammetre.

Weaker responses showed the weakness in candidates' understanding of electrical instruments and their connection with the electrical circuit. In part (a), low-scoring candidates wrote that ammeter should be connected in parallel with the circuit. In part (b), it is necessary to keep its resistance very low to calculate the accurate resistance of an electrical circuit and in part (c), few candidates wrote that a galvanometer can be converted into an ammetre by using a voltmeter in a circuit.

Example:


a.

To measure it very well

b.

To measure the accurate resistance

c.

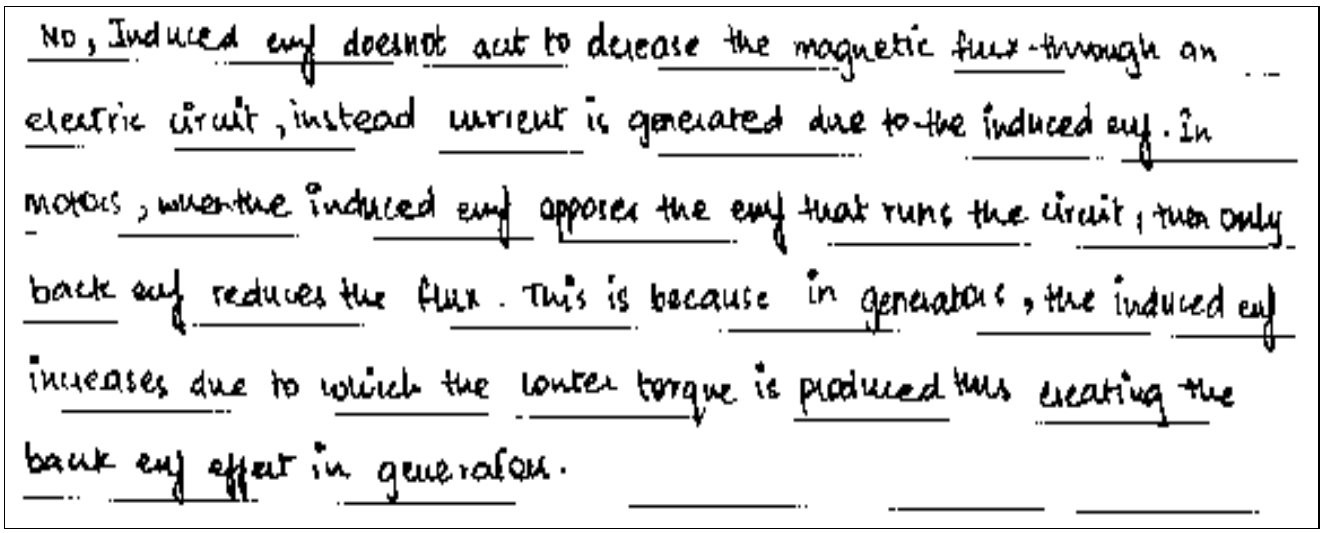
By the help of voltmeter

Question 5:

Does the induced electromotive force (e.m.f.) always act to decrease the magnetic flux through an electric circuit? Explain your answer in TWO points.

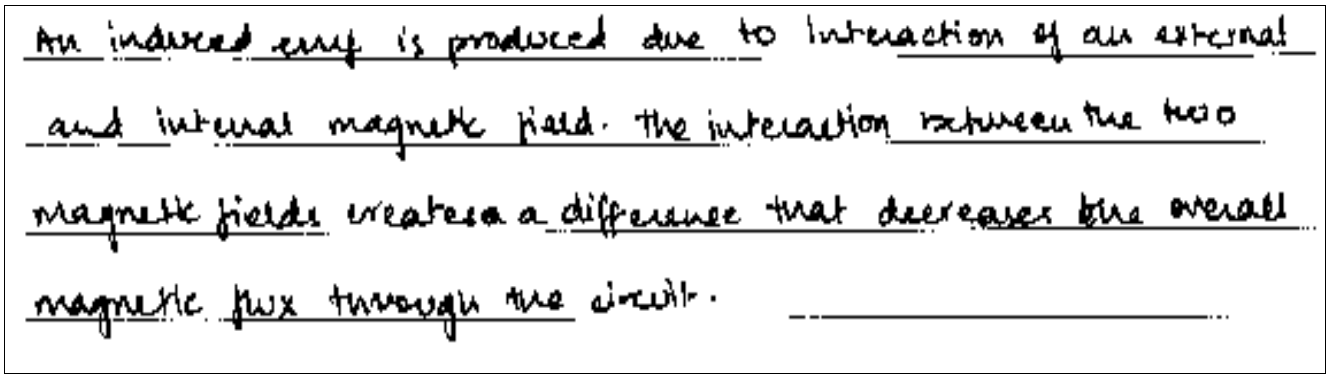
Better responses correctly explained that the induced electromotive force (e.m.f.) does not act to decrease the magnetic flux through an electric circuit by referencing to Lenz's law. High-scoring candidates wrote that the induced e.m.f. always opposes the cause that produces it and if the magnetic flux through the circuit is increased, then the induced e.m.f. acts to decrease the magnetic flux.

Example:



Weaker responses reflected that candidates were unable to explain the given situation. The low-scoring candidates wrote about the concepts of instantaneous value of alternating current, square root of the value of voltage, semi-conductors, internal and external magnetic field and interaction between them which was not relevant to the real demand of the question.

Example:



Question 6:

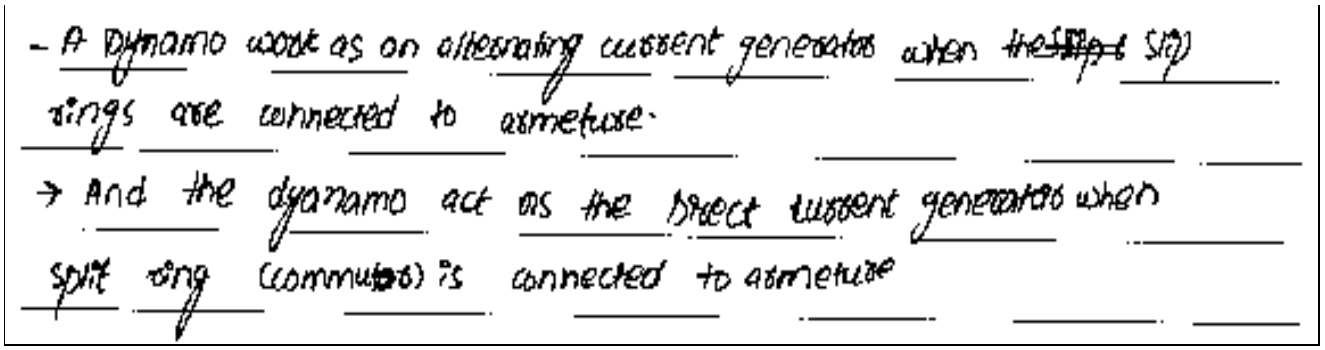
Dynamo is a device that can convert mechanical rotation into electric power.

When does a dynamo work as an alternating current (A.C) generator and a direct current (D.C) generator?

This question was generally well-attempted by the candidates.

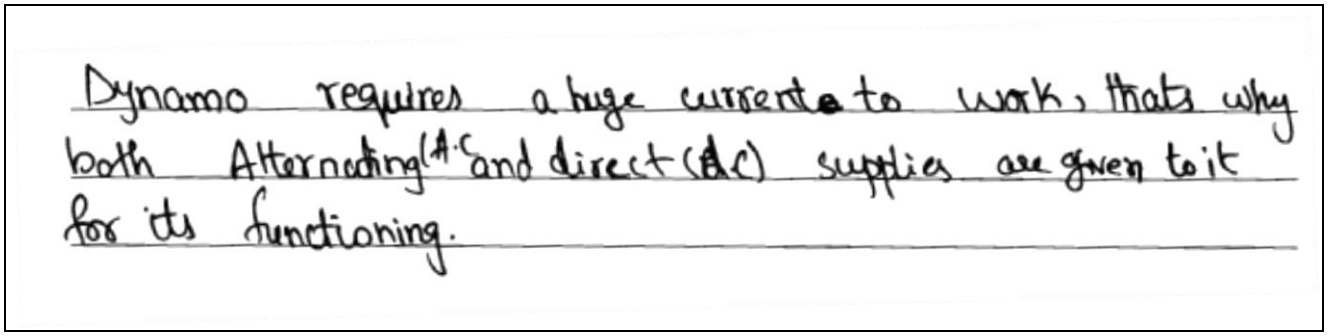
Better responses correctly identified that when a dynamo has slip rings, it works as an alternating current generator and if it has commutator/ spilt rings, it will work as a direct current (DC) generator.

Example:



Weaker responses revealed that candidates were unable to identify the situation in which a dynamo works as an alternating current or a direct current generator. Few of the low-scoring candidates wrote about different characteristics of A.C and D.C current generators.

Example:



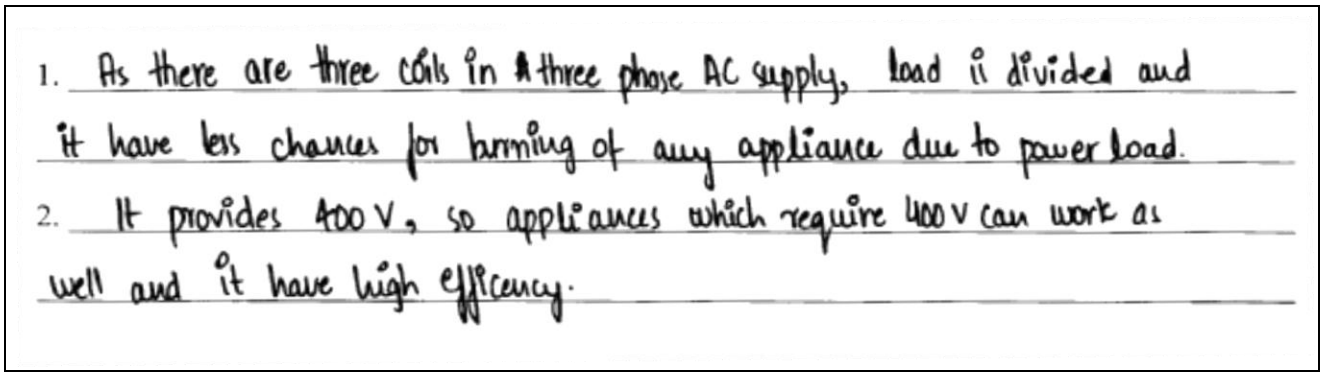
Question 7:

Write any TWO advantages of using three phase alternating current (A.C) supply in a house.

Most candidates attempted this question well.

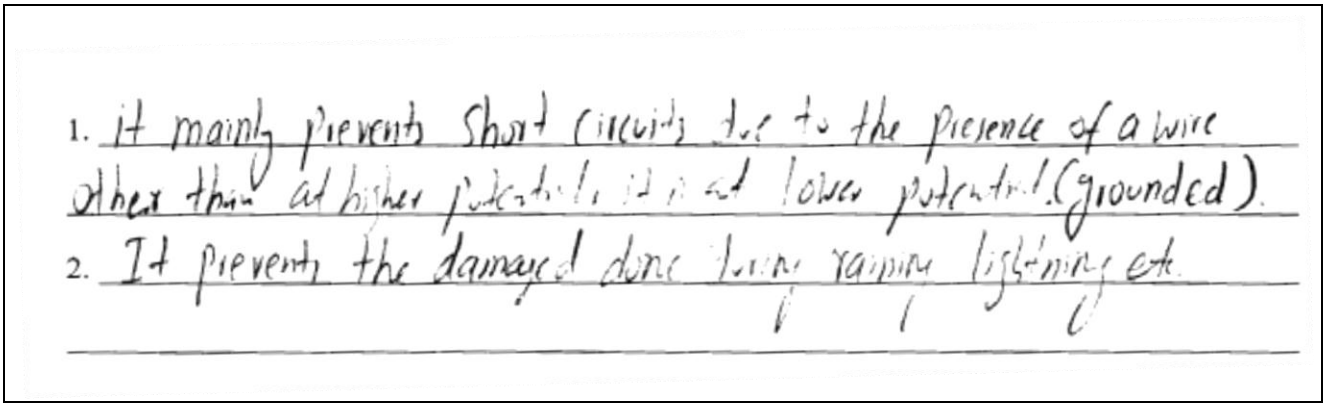
Better responses presented the correct advantages of using three phase alternating current. High-scoring candidates wrote that three phase alternating current can be used to operate some special appliances that require 400 V for their working and can bear heavy power load by dividing the total load of a house into three parts. Some of the candidates wrote that three phase alternating current can divide the load of all the house appliances and in return three phases' voltage does not drop.

Example:


1. As there are three coils in a three phase AC supply, load is divided and it have less chances for burning of any appliance due to power load.
2. It provides 400 V, so appliances which require 400 V can work as well and it have high efficiency.

Weaker responses exhibited that candidates were unable to write two correct advantages. They wrote about short circuit, live and earth wire which were not demanded in the question. Few of the low-scoring candidates repeated the same idea in two different points. Therefore, failed to get full marks.

Example:


1. it mainly prevents short circuits due to the presence of a wire other than at higher potential, it is at lower potential (grounded).
2. It prevents the damages done during raining lightning etc.

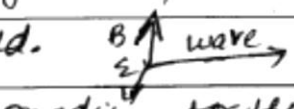
Question 8:

Describe, in THREE points, the production and propagation of electromagnetic waves from a source.

Better responses correctly described the production and propagation of electromagnetic waves by stating that when an alternating potential is applied across the ends of a metallic antenna, an oscillating electric field comes into existence which oscillates the free electrons continuously. High-scoring candidates wrote that the oscillating electrons radiate electromagnetic waves and the changing electric field produces changing magnetic field. Thus, the process continues all along the direction of propagation. Some candidates described that electromagnetic waves change electric and magnetic fields; hence, they are confined to two perpendicular planes and perpendicular to the direction of propagation.

Example:

1. These are produce when charges are subjected to changing electric and magnetic field. A charge when accelarted produces electromagnetic waves. The direction of propagation of it is perpendicular to the magnetic and electric field.
2. Electromagnetic waves ~~are~~ do not need medium for their propagation in space speed is equal to that of light and intensity can be reduced.
3.



Weaker responses were successful in writing only one point which is regarded as fundamental point of the production and propagation of electromagnetic waves, i.e. when an oscillating electric field comes into existence, it oscillates the free electrons and ultimately polarities of the metallic antenna changes after half a cycle. Low-scoring candidates wrote about important components of alpha, beta and gamma particles which were irrelevant to the demand of the question.

Example:

1. The positively charge particles called alpha particles having mass of 4 and 2 proton.
2. The negatively charge particles called Beta particles having no mass.
3. The neutral particles having no mass and no charge called gamma particles.

Question 9:

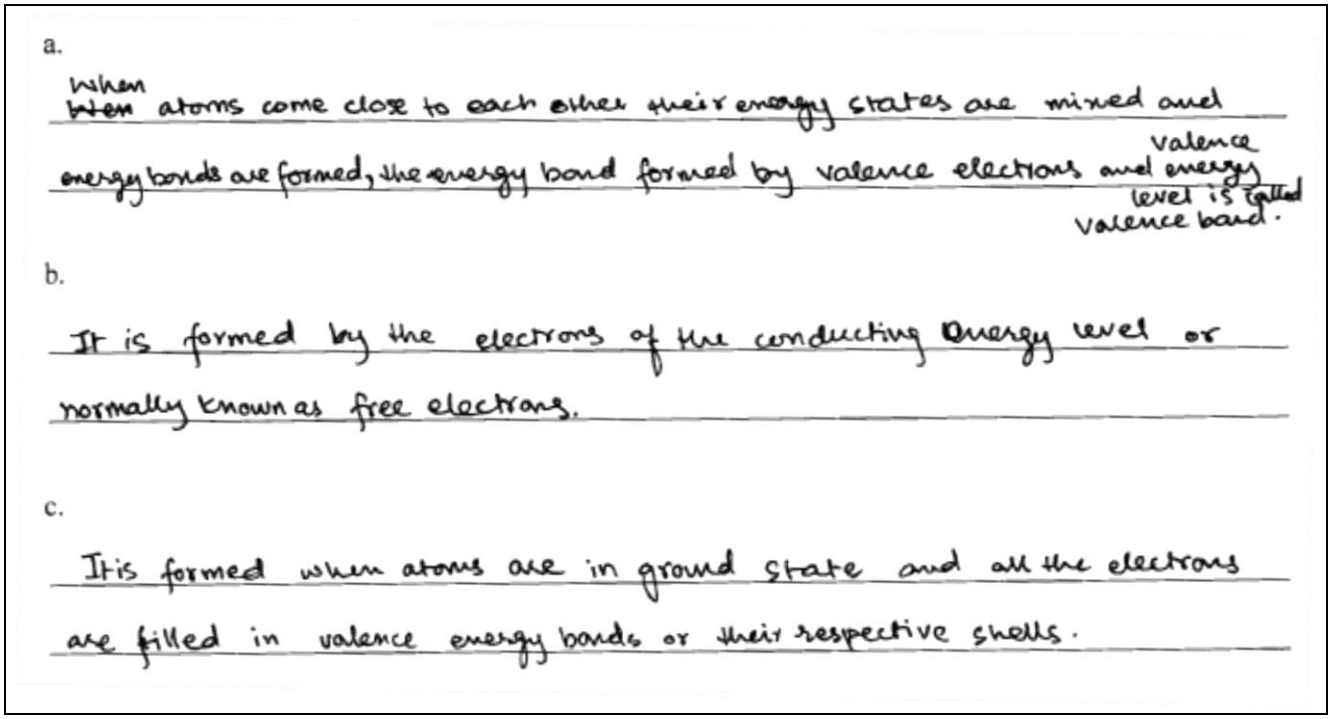
Describe the formation of the following energy bands in solids.

a. Valence energy band
b. Conduction energy band
c. Completely filled energy band

This question was generally well-attempted.

Better responses correctly described the formation of valence energy band, conduction energy band and completely filled energy band. High-scoring candidates described valence energy band as a band that forms in solids due to the overlapping of outermost valence orbits, whereas conduction band is formed in solids due to the overlapping of unoccupied energy levels. Moreover, completely filled energy band in solids is formed by the completely filling of electrons.

Example:

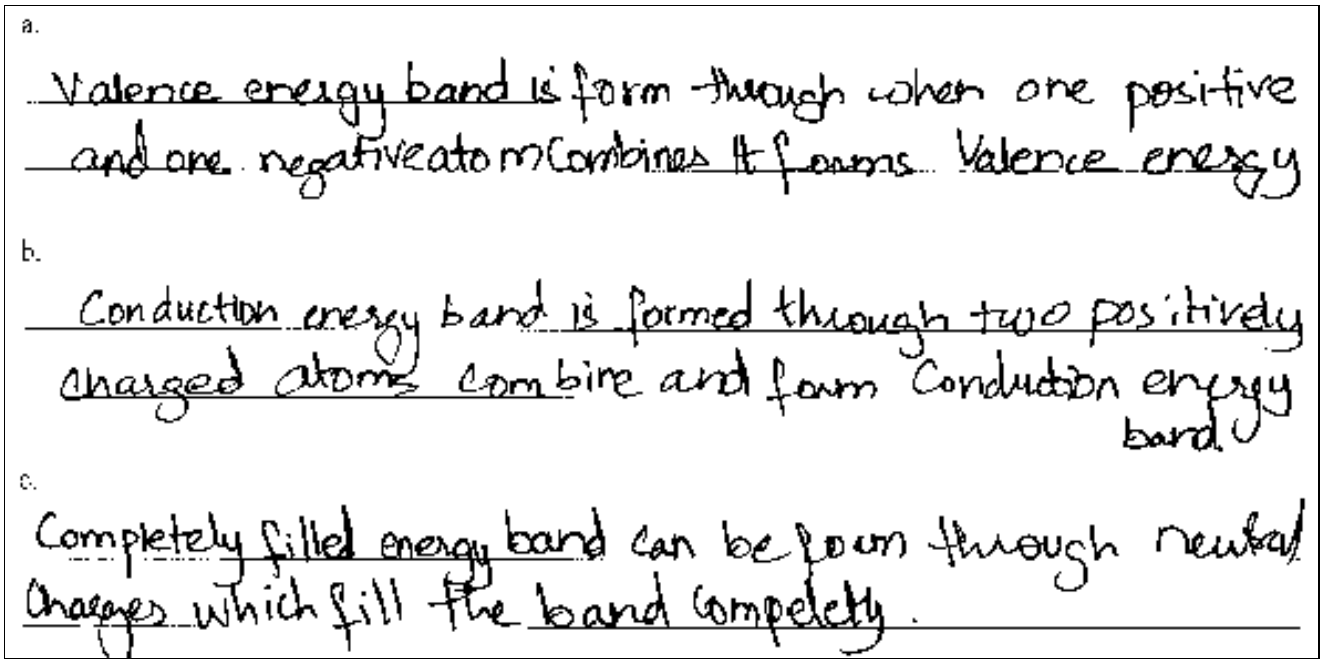

a.
When atoms come close to each other their energy states are mixed and energy bands are formed, the energy band formed by valence electrons and energy level is called valence band.

b.
It is formed by the electrons of the conducting energy level or normally known as free electrons.

c.
It is formed when atoms are in ground state and all the electrons are filled in valence energy bands or their respective shells.

Weaker responses indicated that candidates were unable to describe the formation of the energy bands in solids. Low-scoring candidates wrote about the positive and negative combination of valence energy levels, combination of two positive charged atoms and neutral charges.

Example:


a.

Valence energy band is form through when one positive and one negative atom Combines it forms Valence energy

b.

Conduction energy band is formed through two positively charged atoms combine and form Conduction energy band

c.

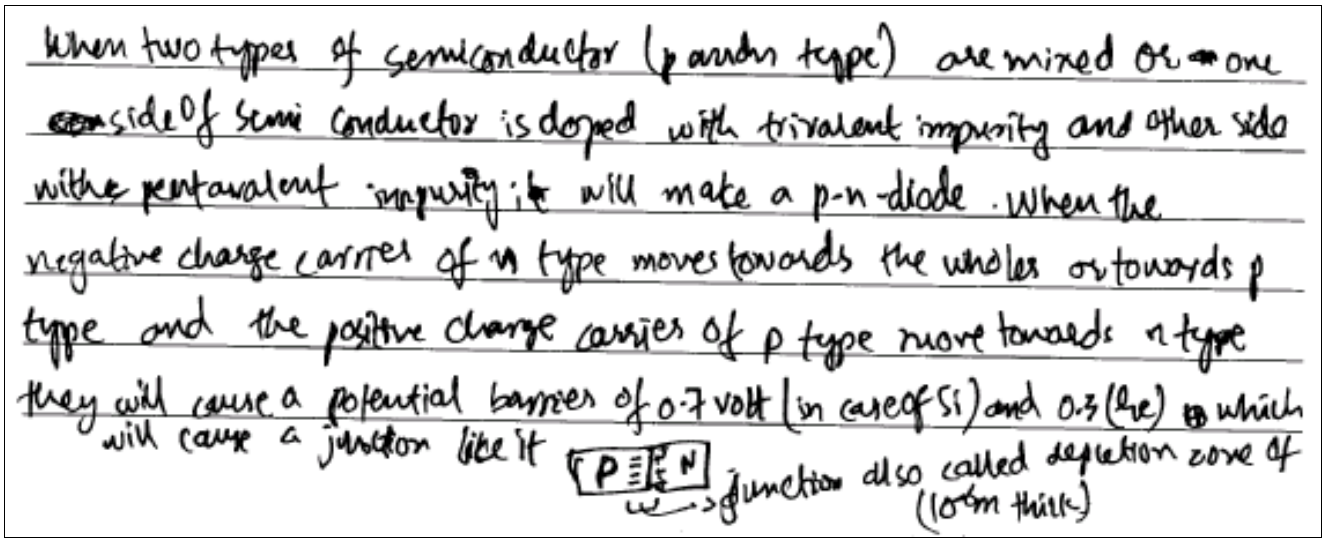
Completely filled energy band can be form through neutral Charges which fill the band completely.

Question 10:

How a depletion region is formed in a p-n junction diode?

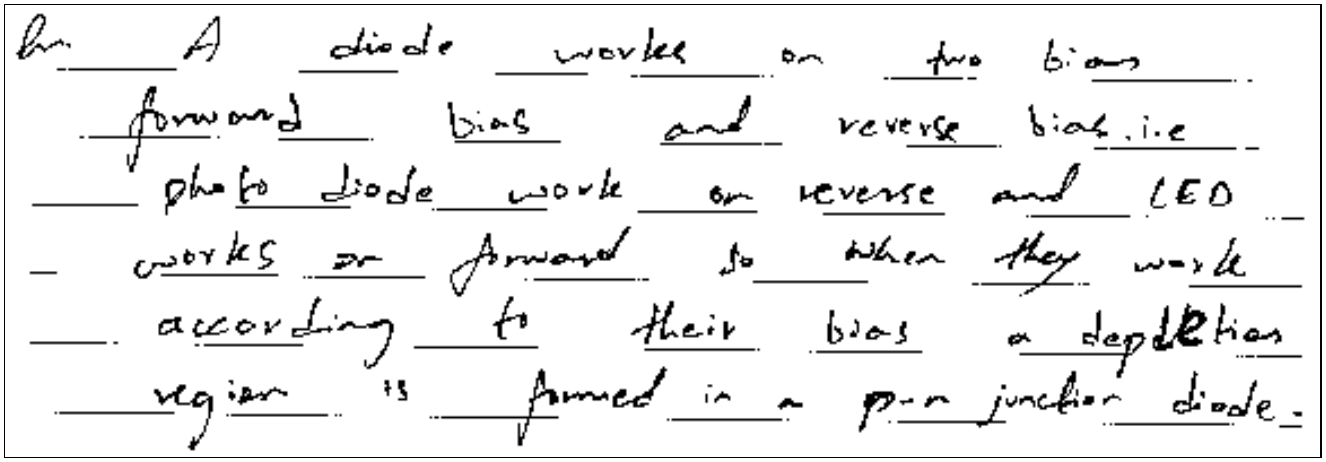
Better responses correctly explained that a depletion region is formed in a p-n junction diode by writing in the following ways that in a p-n junction diode, n-region contains free electrons as the majority charge carriers and p-region contains holes as majority charge carriers. High-scoring candidates wrote that after the formation of the junction, the free electrons in the n-region diffuse into the p-region and as a result of their diffusion finally, the recombination of electrons and holes takes place across the junction that creates a depletion region. Some of the candidates wrote that when the charge carriers are removed, a charge less region is formed around the junction in which charge carriers are not present. Therefore, a depletion region is said to be no net charge region.

Example:


When two types of semiconductor (p and n type) are mixed or one side of Semi Conductor is doped with trivalent impurity and other side with pentavalent impurity it will make a p-n-diode. When the negative charge carrier of n type moves towards the wholes or towards p type and the positive charge carries of p type move towards n type they will cause a potential barriers of 0.7 volt (in case of Si) and 0.3 (Ge) which will cause a junction like it P N → junction also called depletion zone of (10^-6m thick)

Weaker responses wrote the correct biasing phenomena but were unable to explain that how a depletion region is formed in a p-n junction diode. Low-scoring candidates described the concept of forward and reverse biasing and the uses of p-n junction diodes such as in making of LEDs and in solar panels.

Example:



Question 11:

In the special theory of relativity, the Lorentz factor is the factor by which time, length, and mass change for an object moving at speeds close to the speed of light.

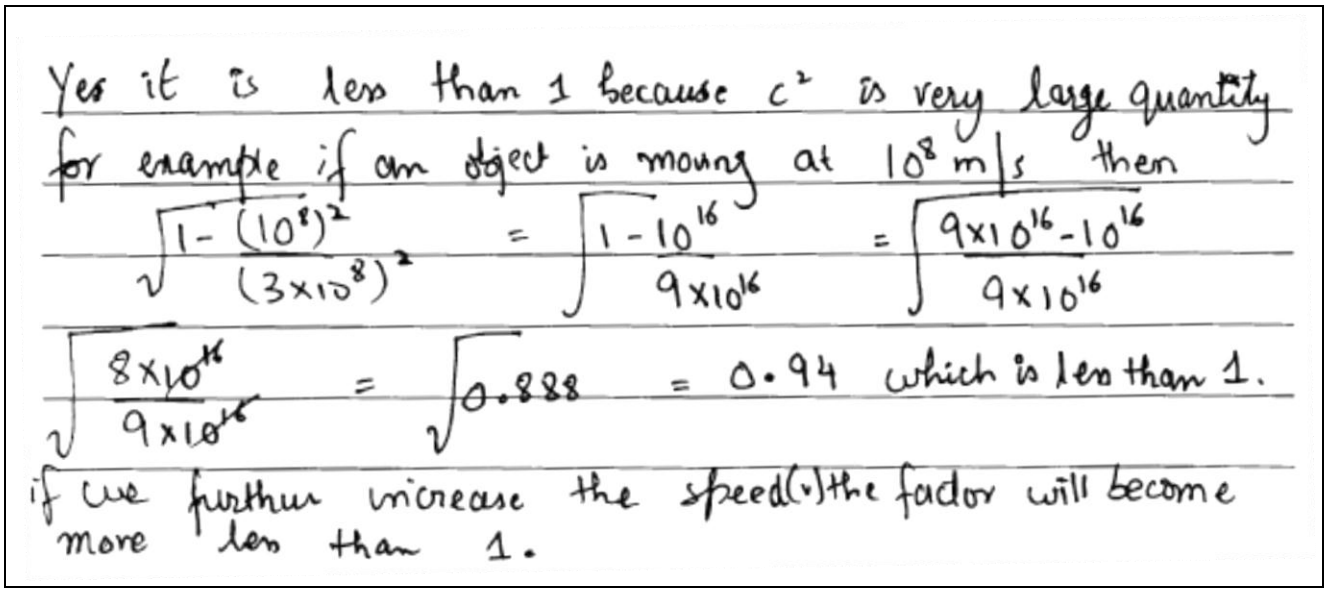
The Lorentz factor is given as $\frac{1}{\sqrt{1-\frac{v^2}{c^2}}}$.

Is the quantity $\sqrt{1-\frac{v^2}{c^2}} < 1$? Justify your answer with the help of mathematical evidence.

Most candidates attempted this question well.

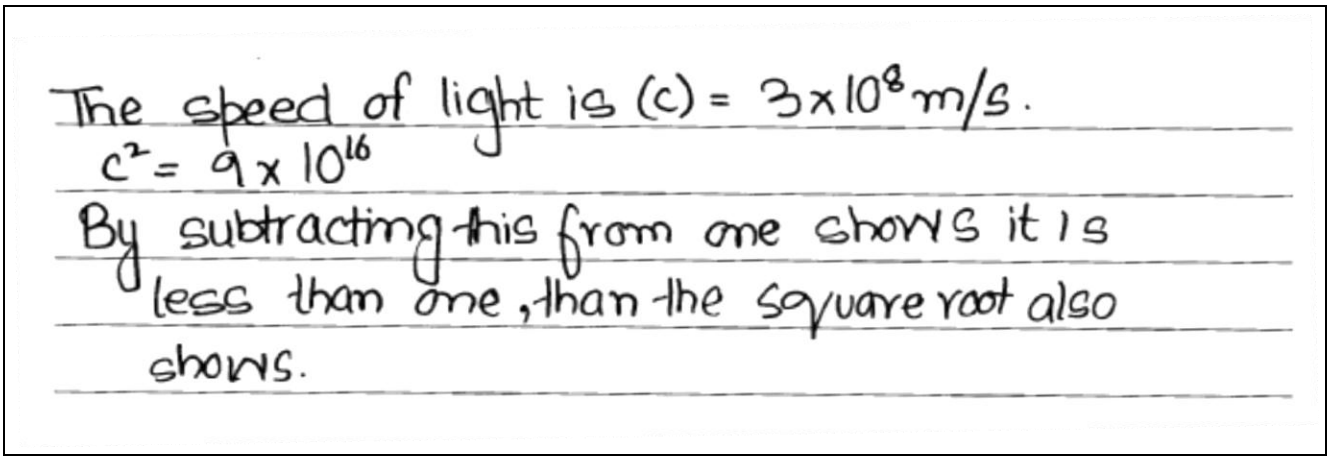
Better responses correctly justified that $\sqrt{1-\frac{v^2}{c^2}} < 1$ with the help of mathematical evidence. High-scoring candidates wrote that no object can attain the speed of light 'c' which means that $\frac{v}{c}$ is always less than one because numerator 'v' is less than the denominator 'c'. Some of the candidates wrote that $\frac{v^2}{c^2}$ is less than one, when $\frac{v^2}{c^2}$ which is less than one is subtracted from one, $1-\frac{v^2}{c^2}$, its result will be less than one, $\left(1-\frac{v^2}{c^2}\right)<1$ and its square root will also be less than one, $\sqrt{1-\frac{v^2}{c^2}}<1$.

Example:



Weaker responses incorrectly justified the Lorentz factor. Few of the low-scoring candidates wrote the speed of light and took out its square root. They were also unable to write all the mathematical evidence to justify their answers that showed lack of in-depth understanding of the candidates on this very important topic of special theory of relativity.

Example:


The speed of light is (c) = 3×10^8 m/s.
$c^2 = 9 \times 10^{16}$
By subtracting this from one shows it is less than one, than the square root also shows.

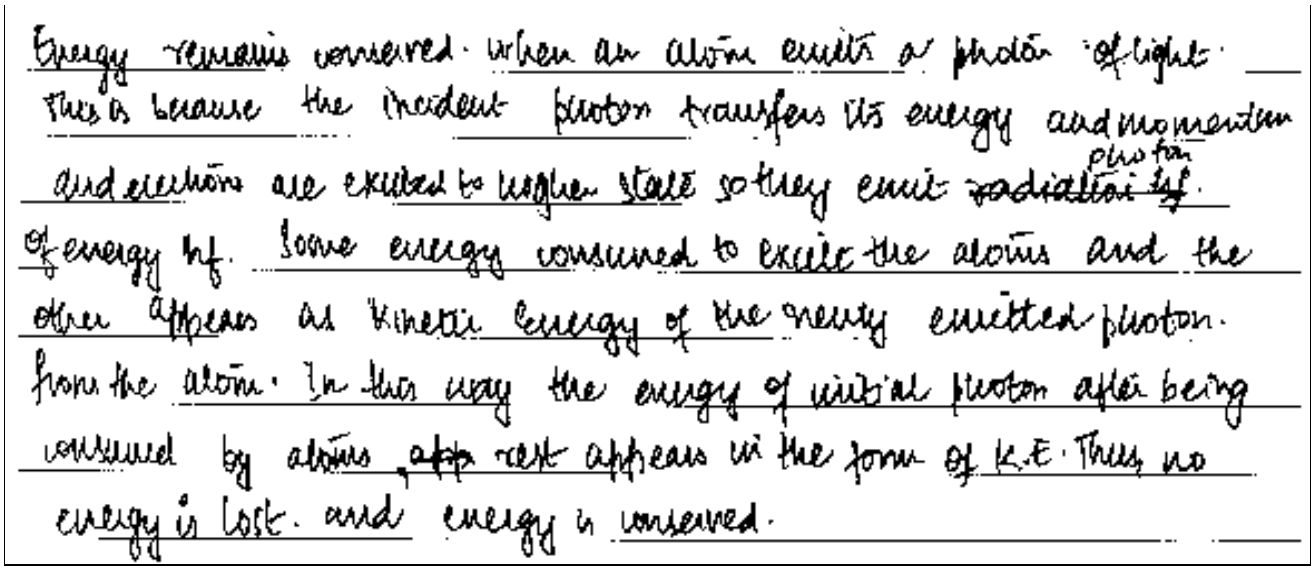
Question 12:

Is energy conserved when an atom emits a photon of light? Justify your answer.

Most candidates attempted this question well.

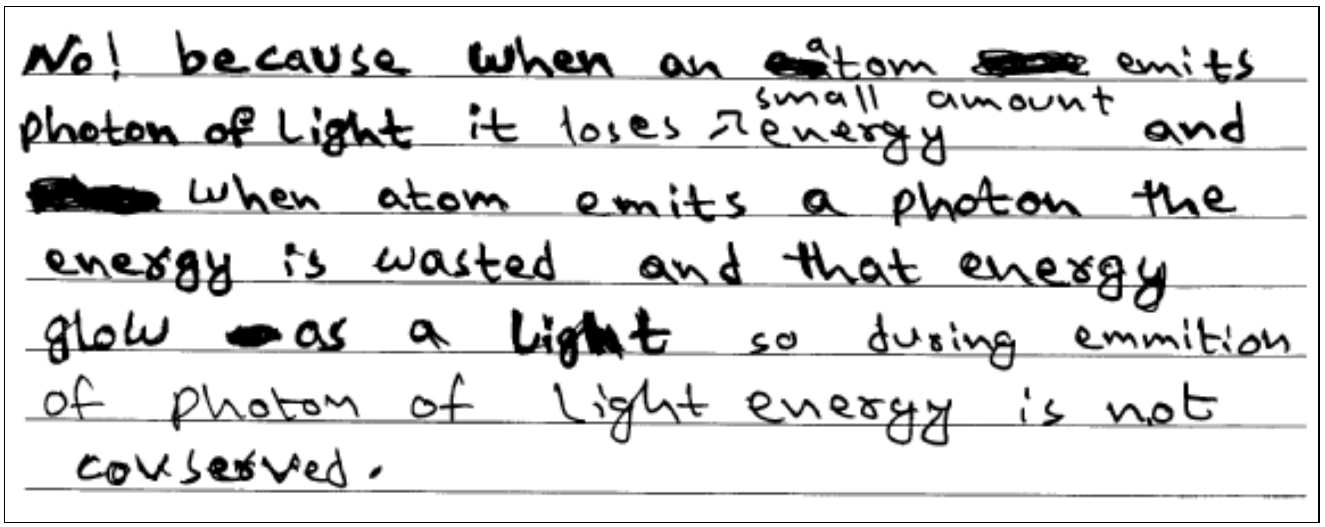
Better responses correctly justified that energy is conserved when an atom emits a photon of light by stating that when an atom is excited then energy is supplied to the electron by some external source. The same energy is emitted in the form of light photon when it returns back to its ground state. Some of the candidates wrote that the energy absorbed by the atom during excitation exactly equals to the energy emitted during its de-excitation. Thus, law of energy is conserved, i.e. total energy remains the same.

Example:


Energy remains conserved. When an atom emits a photon of light. This is because the incident photon transfers its energy and momentum and electrons are excited to higher state so they emit radiation plus fm of energy hf. Some energy consumed to excite the atoms and the other appears as kinetic energy of the newly emitted photon from the atom. In this way the energy of initial photon after being consumed by atoms, at rest appears in the form of K.E. Thus no energy is lost. and energy is conserved.

Weaker responses incorrectly justified the phenomenon. Few of the low-scoring candidates wrote that when an atom emits a photon of light, it wasted the energy in glowing and emitting photon that showed lack of in-depth understanding of the candidates on this very important topic of atomic spectra.

Example:



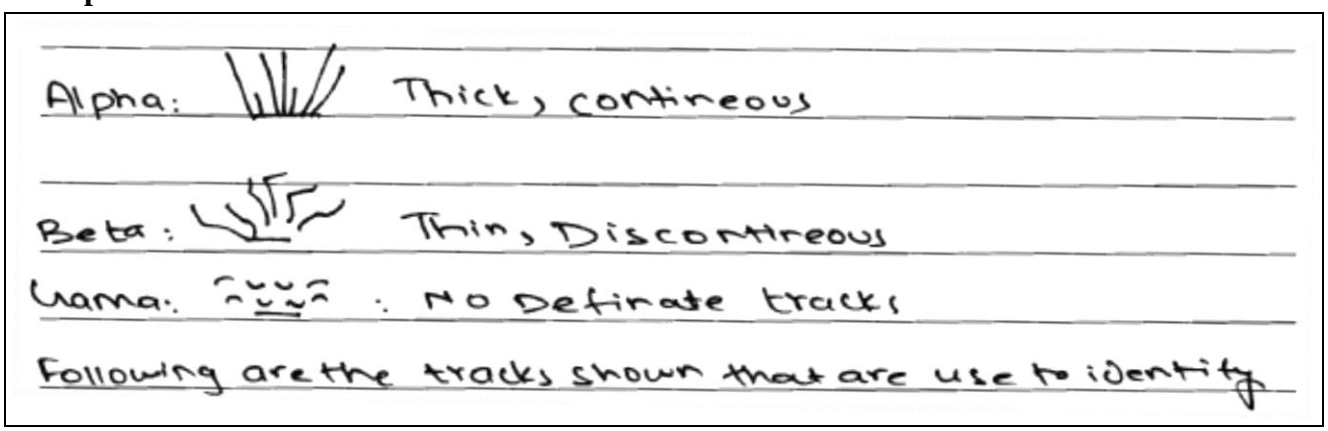
Question 13:

How does the length and shape of the tracks of a radioactive incident particle in Wilson cloud chamber indicate the nature of particle, i.e. alpha, beta or gamma?

Most candidates attempted this question well.

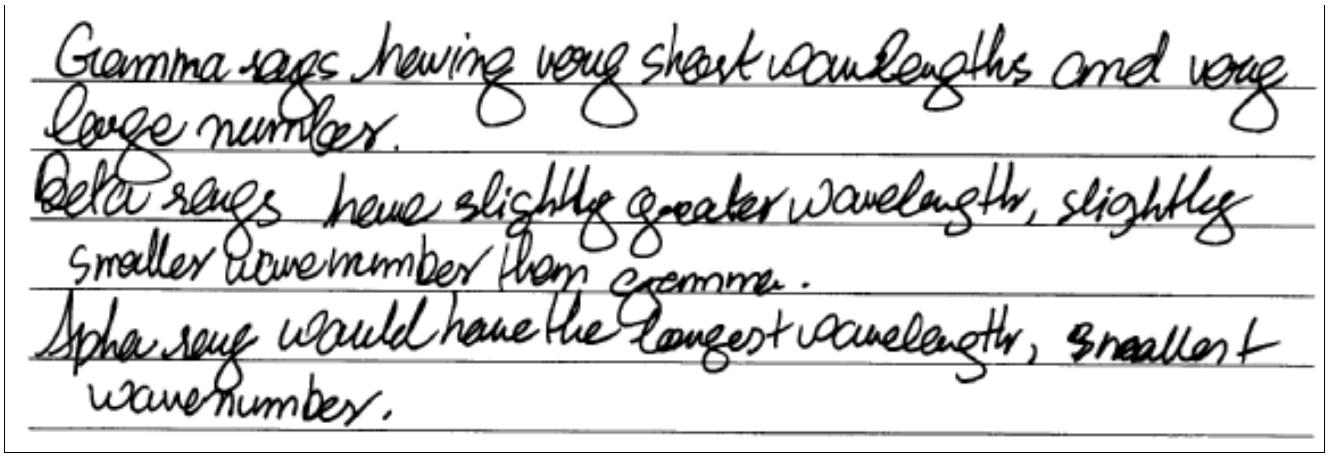
Better responses correctly described the length and shape of the tracks of a radioactive incident particle in Wilson cloud chamber to indicate the nature of particle, i.e. alpha, beta or gamma. High-scoring candidates wrote that the alpha particle is represented by straight, thicker and continuous tracks because it has larger mass and greater ionising power; beta particle is represented by thinner, short and discontinuous tracks because these particles have less mass and less value of ionising power, whereas Gamma rays leave behind no definite tracks along their path because of high penetrating power and very small ionising power.

Example:



Weaker responses incorrectly described the length and shape of the tracks of a radioactive incident particle in Wilson cloud chamber. Few of the low-scoring candidates wrote about short and long wavelength and wave numbers that showed lack of in-depth understanding of the candidates on this very important topic of nuclear physics.

Example:


Gamma rays hewing very short wavelengths and very large number.
Beta rays have slightly greater wavelength, slightly smaller wavenumber than gamma.
Alpha rays would have the longest wavelength, smallest wavenumber.

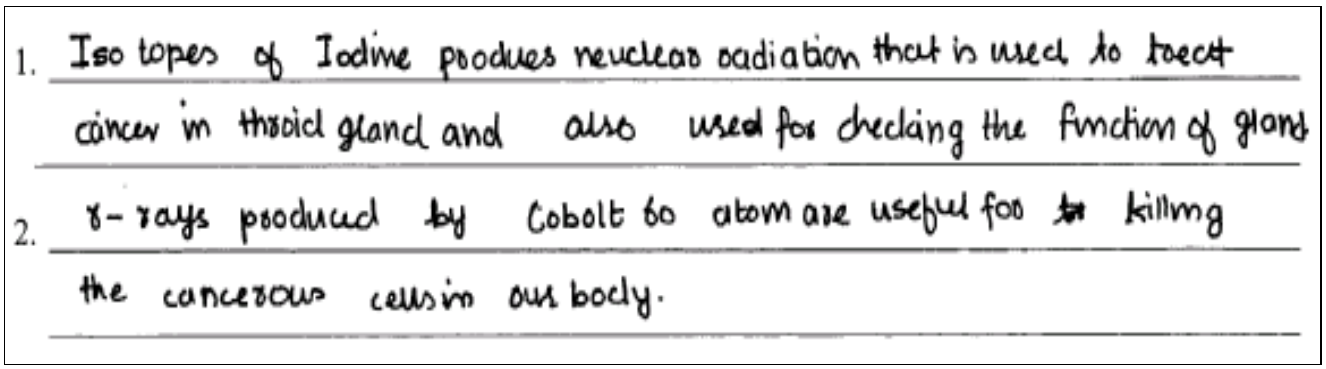
Question 14:

Write an example each of medical diagnosis and treatment carried out by nuclear radiation.

Most candidates attempted this question well.

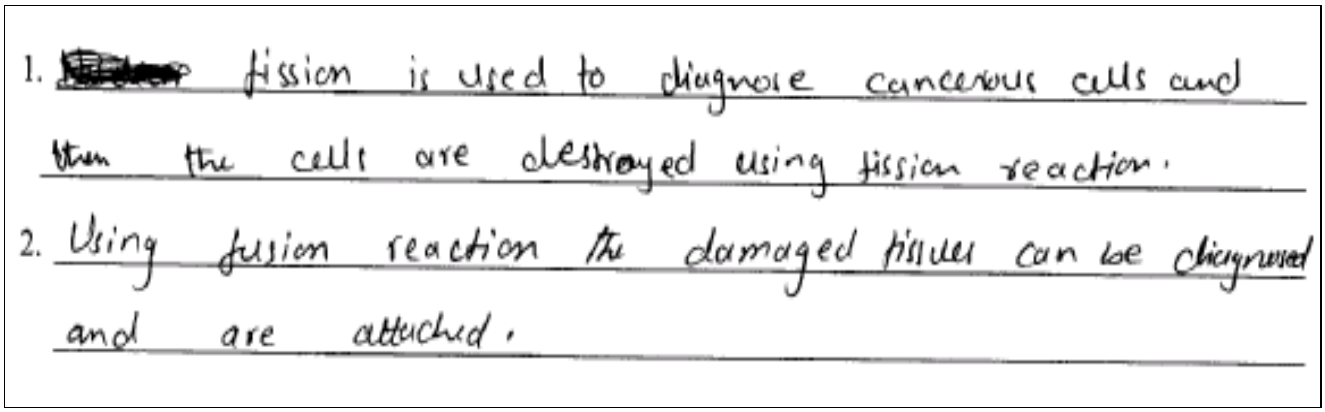
Better responses correctly wrote an example each of medical diagnosis and treatment carried out by nuclear radiation. High-scoring candidates wrote that radio-isotopes are used to cure tumour/ cancer, leukaemia and for the treatment of heart diseases/ used to diagnose tumours in brain and other parts of the human body/ used in X-rays to see the condition of bones. Some of the candidates wrote about the use of radioactive iodine to treat thyroid problems and thyroid cancer/ gamma rays from cobalt-60 are used for cancer treatment.

Example:


1. Iso topes of Iodine produes neuclear radiation that is used to treat cancer in throid gland and also used for checking the function of gland
2. γ- rays produced by Cobolt 60 atom are useful for the killing the cancerous cells in our body.

Weaker responses wrote incorrect examples each of medical diagnosis and treatment carried out by nuclear radiation. Few of the low-scoring candidates wrote about the cancerous cells which are destroyed by the nuclear fission reaction.

Example:


1. [redacted] fission is used to diagnose cancerous cells and then the cells are destroyed using fission reaction.
2. Using fusion reaction the damaged tissues can be diagnosed and are attached.

Extended Response Questions (ERQs)

These questions offered a choice between part **a** and **b**

Majority of the candidates attempted Question **15b** and they scored well in this part.

Question 15a:

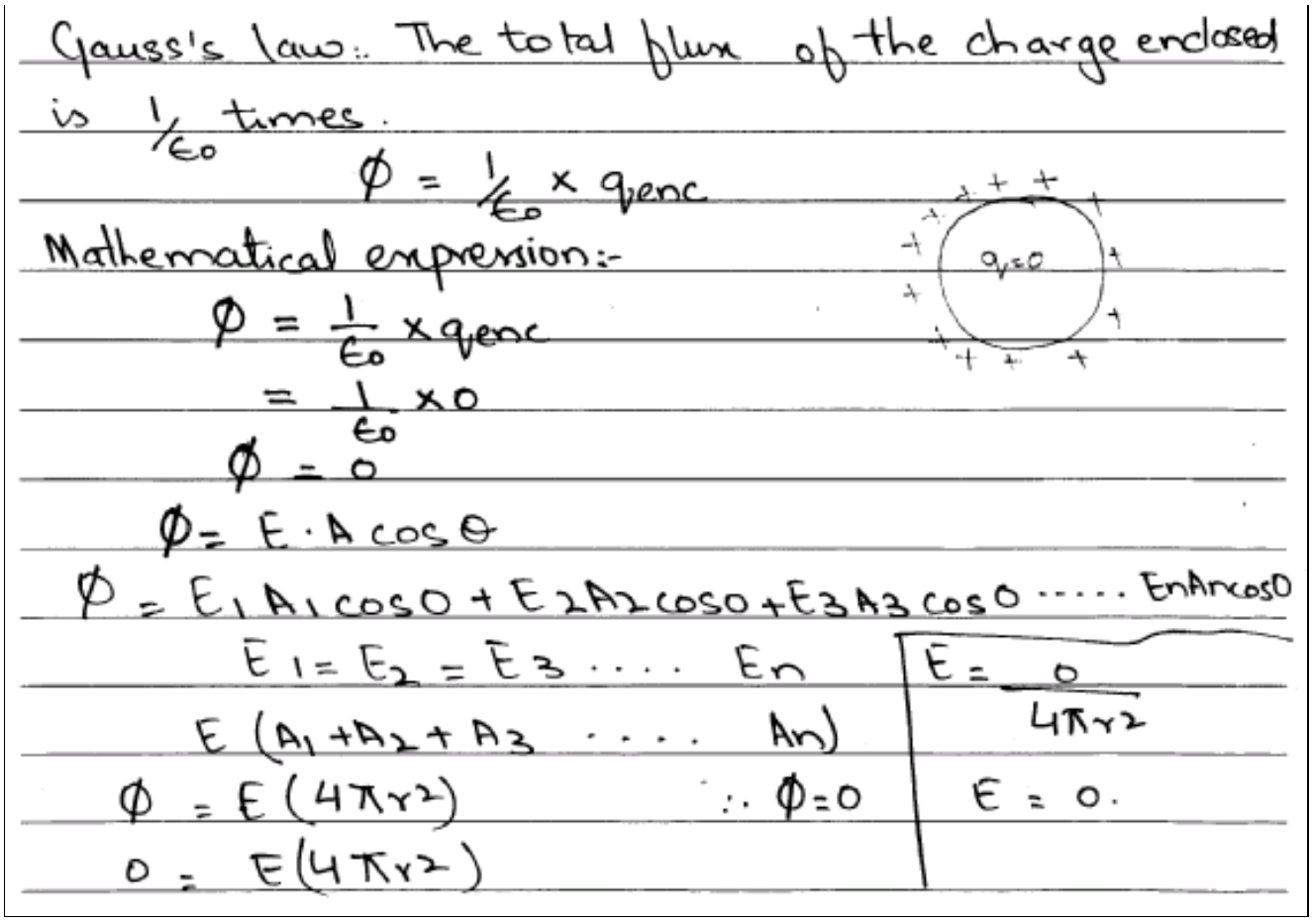
i. State Gauss' law.

ii. Is it necessary that electric field intensity $\vec{E}$ remains zero inside a charged spherical rubber balloon? Justify your answer with the help of a mathematical evidence.

(**Note**: Assuming that charges are distributed uniformly over the surface.)

Better responses correctly stated the Gauss' law and justified that electric field intensity remains zero inside a charged spherical rubber balloon. In part (a), high-scoring candidates wrote that the total flux through a closed surface is $\frac{1}{\varepsilon_o}$ times that are charges enclosed in a surface. In part (b), candidates wrote that if the Gaussian surface is imagined inside a charged balloon, then it does not contain any charge i.e. q = 0, $\varphi c = \frac{q}{\varepsilon o} = 0$, $\varphi c = \vec{E}(4\pi r^2)$, $0 = \vec{E}(4\pi r^2)$ and E = 0.

Example:



Weaker responses incorrectly stated the Gauss' law and failed to justify that electric field intensity remains zero inside a charged spherical rubber balloon. Some of the low-scoring candidates made the correct imaginary diagram but were unable to justify their answers with the help of mathematical evidence. Few of the candidates expressed electric field and electric field intensity in their answers which were irrelevant to the demand of the question.

Example:

Part (a)

(i) Gauss's Law:- Gauss's Law states that the electric flux through any closed surface is always equal to the number of charges enclosed in it.

(ii) ~~Yes~~ No, it is not necessary that the electric field intensity $\vec{E}$ remains zero inside a spherically charged rubber balloon although the charges are distributed uniformly over the surface of the balloon, so the charges will only exist on the surface of the balloon and not a single charge exists within the rubber balloon, so it means that when no charges are present within this balloon so no electric field will exist within it, ~~that can cause an electric field intensity~~ but it does not happen as we can't do mathematically, electric field intensity is given as '$\varepsilon = F/q_o$', when we are interested in finding the electric field intensity within this balloon we put $q_o = 0$ (as no charge exists) then we get $\varepsilon = \frac{F}{0} = \infty$. The electric field intensity within this region is infinite.

Question 15b:

A parallel plate capacitor is connected to a battery.

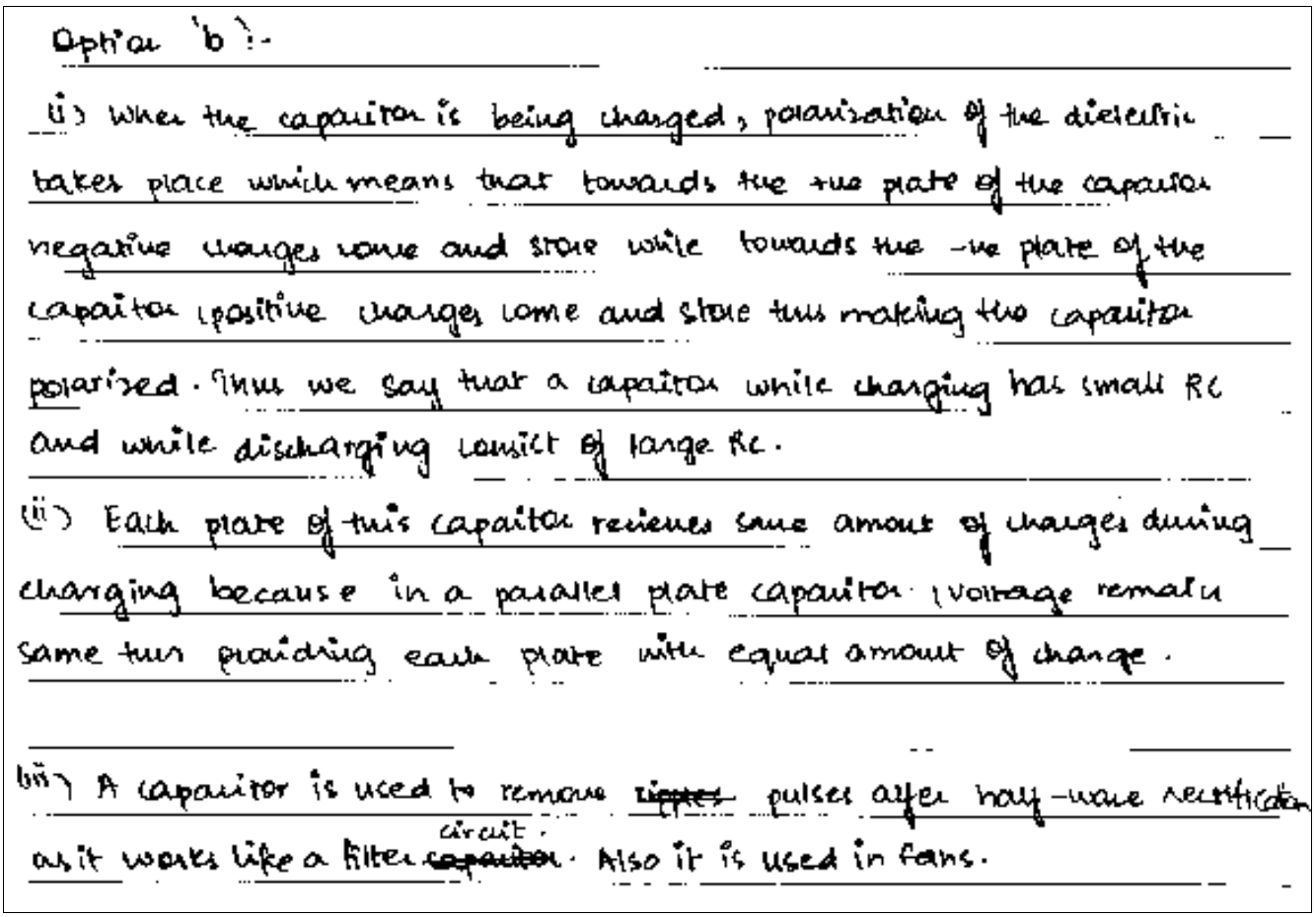
i. Describe the process taking place at the two terminals of the battery when a capacitor is being fully charged.

ii. Each plate of this capacitor receives same amount of charges during charging. Give a reason for this statement.

iii. Write ONE function of a capacitor.

Better responses correctly described the process taking place at the two terminals of the battery when a capacitor is being fully charged and gave the correct reason that each plate of this capacitor receives same amount of charges during charging alongwith one correct function of a capacitor. In part (i), high-scoring candidates described that when the plates of a capacitor are connected to the terminals of a battery, then the plate that is connected to the negative terminal of the battery become negatively charged. Some of the candidates wrote that positive charges will pass through the battery and will be deposited on the other plate until this process will continue till the potential difference between the plates becomes equal to the electromotive force (e.m.f.) of the battery and at this point the capacitor will be fully charged. In part (ii), candidates wrote the reason that when a parallel plate capacitor is connected across a battery, the electrons transferred from a positive plate are exactly equal to the electrons received by the opposite negative plate and in part (iii), candidates wrote that capacitors are used in memory, flash light, resonance circuit and for storing energy in electrostatic field.

Example:



Option 'b':-

(i) When the capacitor is being charged, polarisation of the dielectric takes place which means that towards the +ve plate of the capacitor negative charges come and store while towards the -ve plate of the capacitor (positive charges come and store thus making the capacitor polarised. Thus we say that a capacitor while charging has small RC and while discharging consist of large RC.

(ii) Each plate of this capacitor recieves same amount of charges during charging because in a parallel plate capacitor (voltage remain same thus providing each plate with equal amount of charge.

(iii) A capacitor is used to remove ~~ripples~~ pulses after half-wave rectification as it works like a filter ~~capacitor~~ circuit. Also it is used in fans.

Weaker responses reflected that candidates were unable to understand the question. Some of the low-scoring candidates showed in their description of part (i) the derivation of capacitance of a parallel plate capacitor. But few of the candidates wrote the correct answers in part (ii) and (iii) regarding the amount of charges and function of capacitor.

Example:

(b)

(i) initially the charge on capacitor is 0. but when we start charging it becomes V.

$$\frac{0+V}{2} = \frac{V}{2}$$

So energy on single terminal of battery.

$$\mathcal{E} = qV$$

$$\mathcal{E} = \frac{qV}{2} \qquad q = CV$$

$$\mathcal{E} = \frac{CV^2}{2}$$

So on both terminals $\mathcal{E} = \frac{CV_1^2}{2} + \frac{CV_2^2}{2}$

(ii) Due to potential difference b/w the plates charge on each capacitor will be same. $\mathcal{E} = \frac{CV_1^2}{2} + \frac{CV_2^2}{2}$

(iii) Capacitors are used to store charge b/w metallic plates.

Majority of the candidates attempted Question **16b** and they scored well in this part.

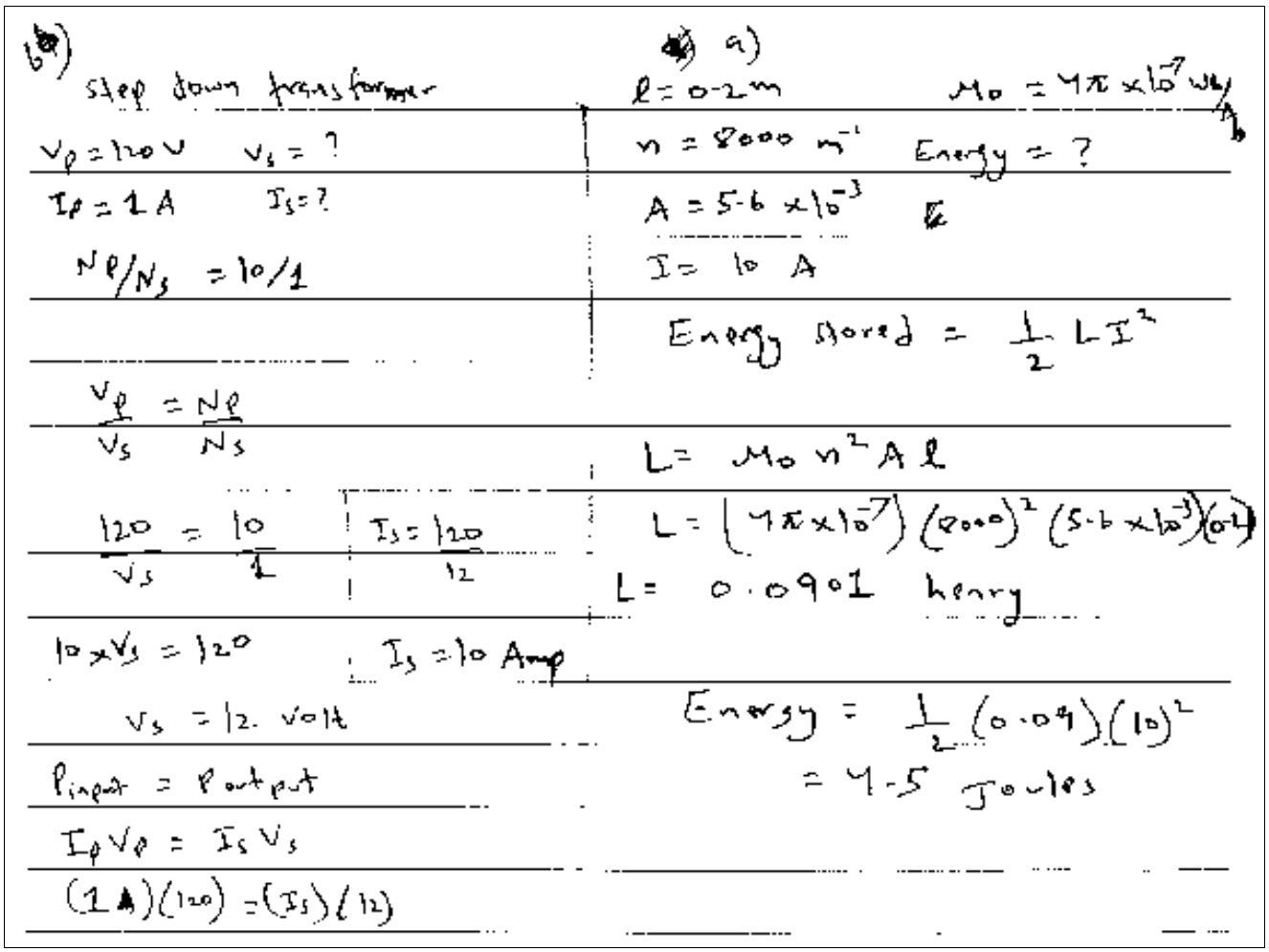
Question 16a:

A 0.2 m long inductor has 8000 turns per metre and its cross-sectional area is 5.6×10^{-3} m^2. If 10 A current passes through the inductor, then calculate the value of energy stored in it.

(**Note**: The value of $\mu_o = 4\pi \times 10^{-7}$ T.m/A)

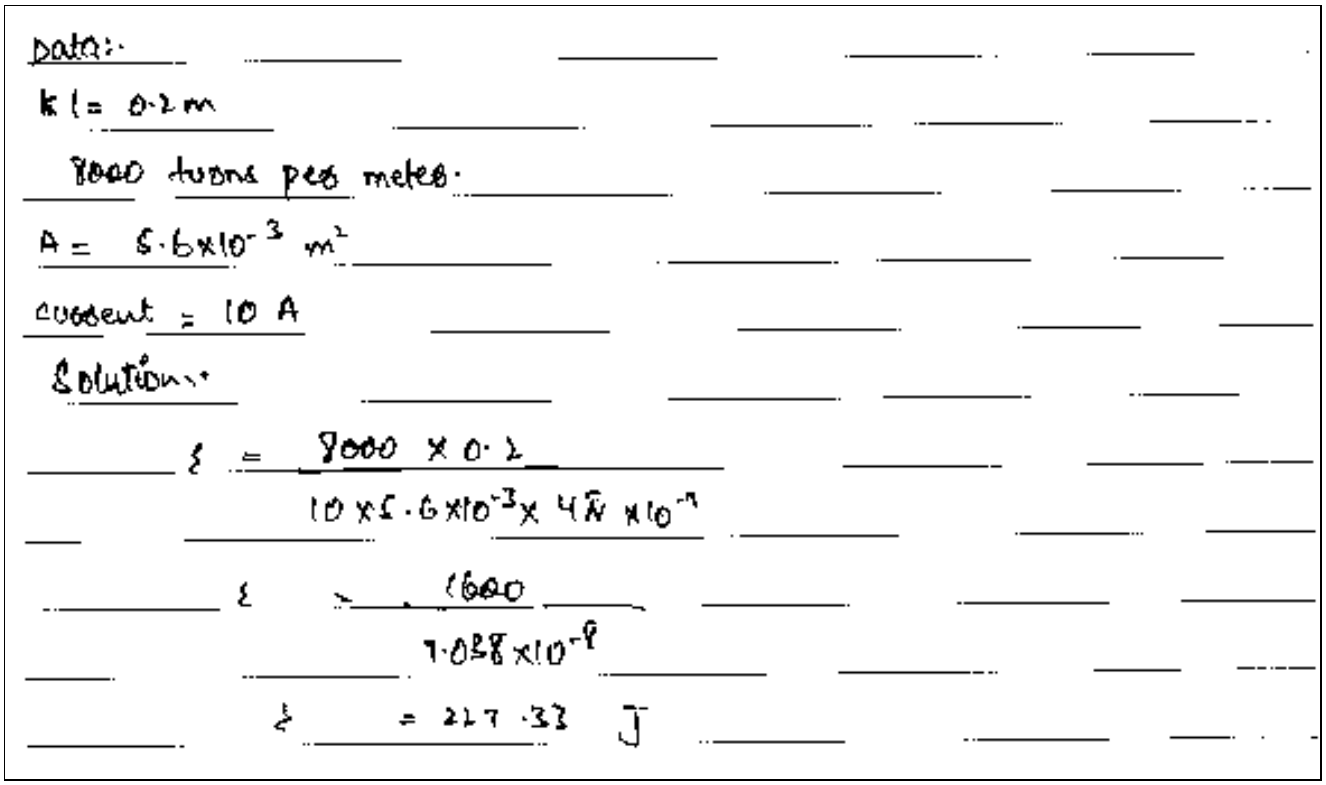
Better responses correctly calculated the value of energy stored in an inductor. High-scoring candidates extracted the correct data, used the correct formula $E = \frac{1}{2} L I^2$ and $L = \mu_o n^2 A l$ and calculated the correct answer. Few of the candidates attempted both the parts and calculated the correct answers that showed their good understanding and command over the topics.

Example:



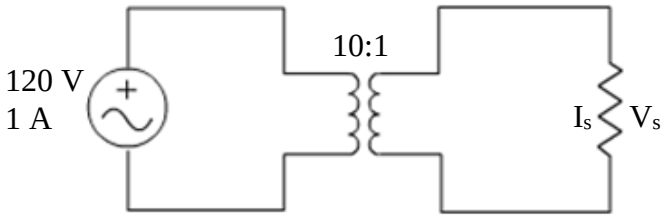
Weaker responses showed candidates' weak knowledge about induction. Most of the low-scoring candidates were unable to write neither the correct data nor the formula of energy stored in an inductor; hence, failed to calculate the correct answer.

Example:



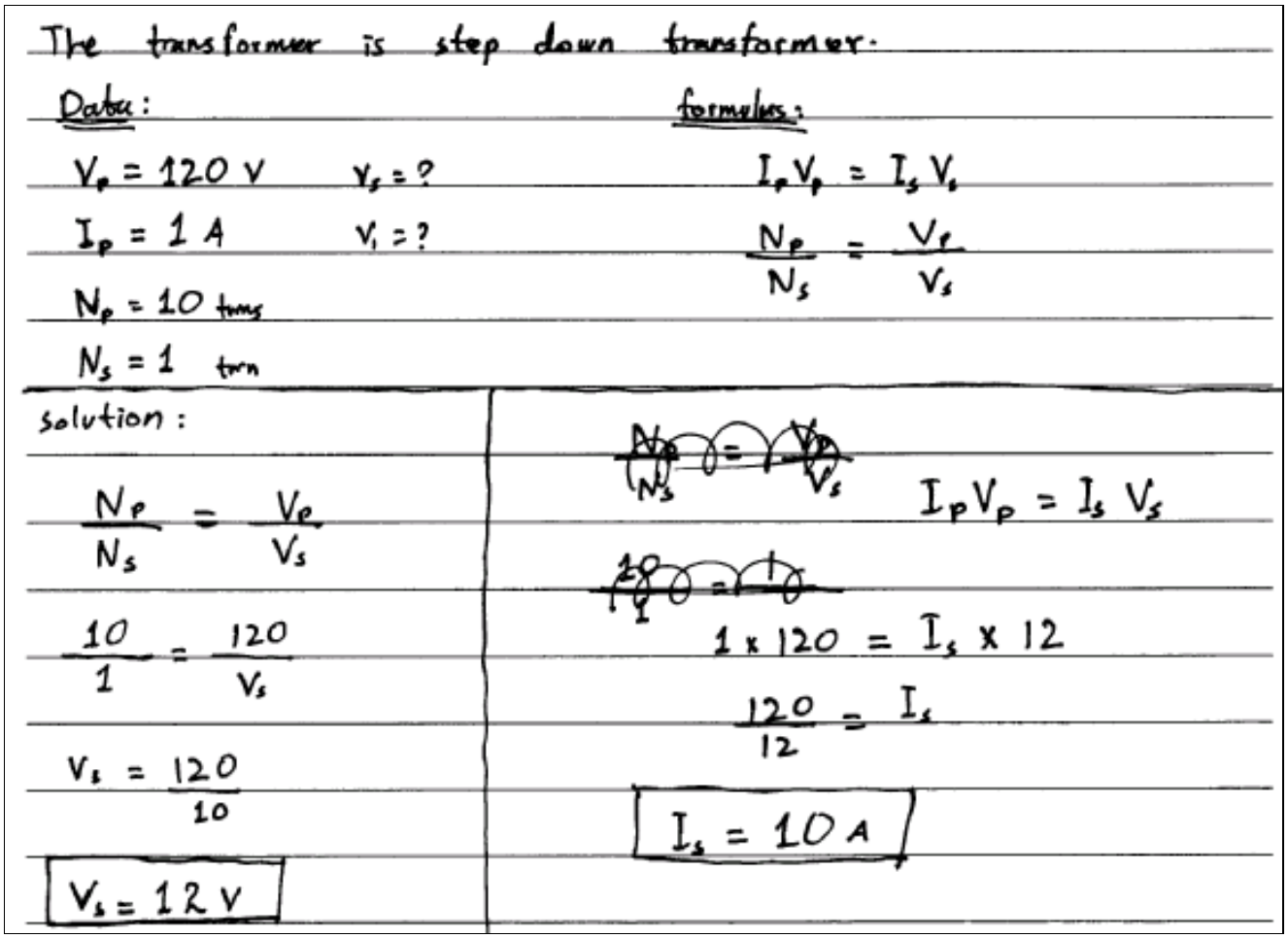
Question 16b:

In the given circuit diagram, determine the voltage and current at the secondary coil of a transformer.



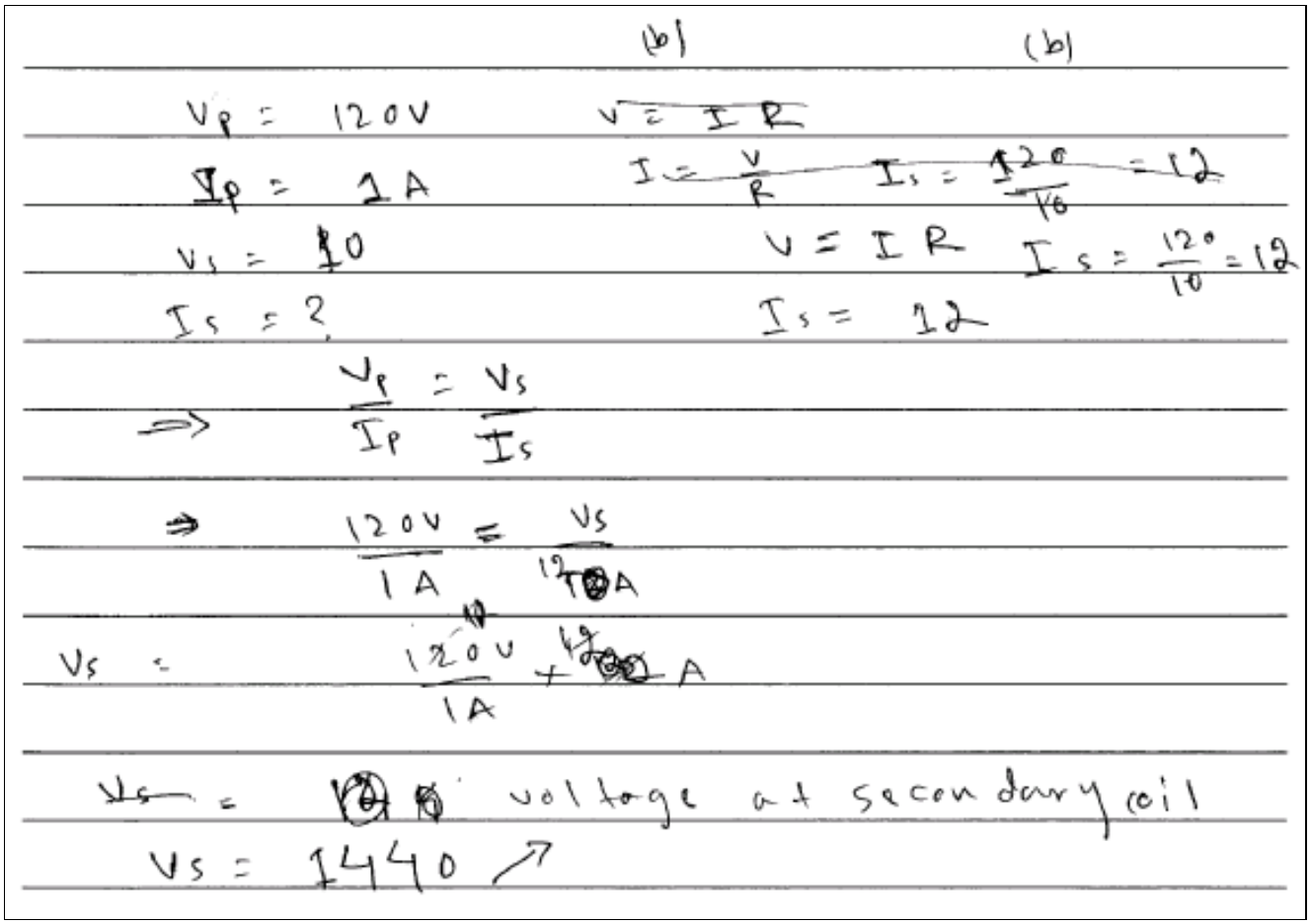
Better responses correctly determined the voltage and current at the secondary coil of a transformer with the help of the given circuit diagram in the question. High-scoring candidates extracted the correct data, used the correct formula of voltage ratio and current ratio $\frac{V_p}{V_s} = \frac{N_p}{N_s}$ and $\frac{I_s}{I_p} = \frac{N_p}{N_s}$ (OR) $\frac{I_s}{I_p} = \frac{V_p}{V_s}$ respectively and hence, calculated the correct answers.

Example:



Weaker responses were successful in calculating either the voltage or current at the secondary coil but failed to calculate both parts correctly. Few of the low-scoring candidates used Ohm's law for calculating the value of current but were unable to get the correct answer. The major flaw was that the candidates incorrectly used the correct formula of voltage ratio.

Example:


(b)
(b)
$V_p = 120V$
$V = IR$
$I = \frac{V}{R}$
$I_s = \frac{120}{10} = 12$
$I_p = 1A$
$V_s = 10$
$V = IR$
$I_s = \frac{120}{10} = 12$
$I_s = ?$
$I_s = 12$
$\Rightarrow \frac{V_p}{I_p} = \frac{V_s}{I_s}$
$\Rightarrow \frac{120V}{1A} = \frac{V_s}{12A}$
$V_s = \frac{120V}{1A} \times 12A$
$V_s =$ voltage at secondary coil
$V_s = 1440$

Majority of the candidates attempted Question **17a** and they scored well in this part.

Question 17a:

In light of Plank's hypothesis, which one of the photon corresponding to red, green and blue light carries

i. the most energy.
ii. the most momentum.

Explain your answer by providing mathematical evidence.

Better responses correctly explained their findings in light of Plank's hypothesis about the photon corresponding to red, green and blue light carrying the most energy and momentum with mathematical evidence. In part (i), candidates wrote E = hf, $f = \frac{c}{\lambda}$, $E = \frac{hc}{\lambda}$, $E = (\text{constant}) \times \frac{1}{\lambda}$ and $E \alpha \frac{1}{\lambda}$. In part (ii), candidates explained as $p = \frac{h}{\lambda}$, $p = (\text{constant}) \times \frac{1}{\lambda}$ and $p \alpha \frac{1}{\lambda}$.

Example:

Photo electric effect :- When the electrons are emitted out from a metal surface, it is known as photoelectric effects.

•) The electrons are only emitted from cathode when light falls on it. The photoelectric current stops flowing as soon as the incident light is stopped.

•) The maximum energy of electrons can be calculated by reversing the connections of electrode with terminal of battery. However, at some time, the current will become zero at V_0 (stopping potential).

$\frac{1}{2} mv^2 = V_0 e$

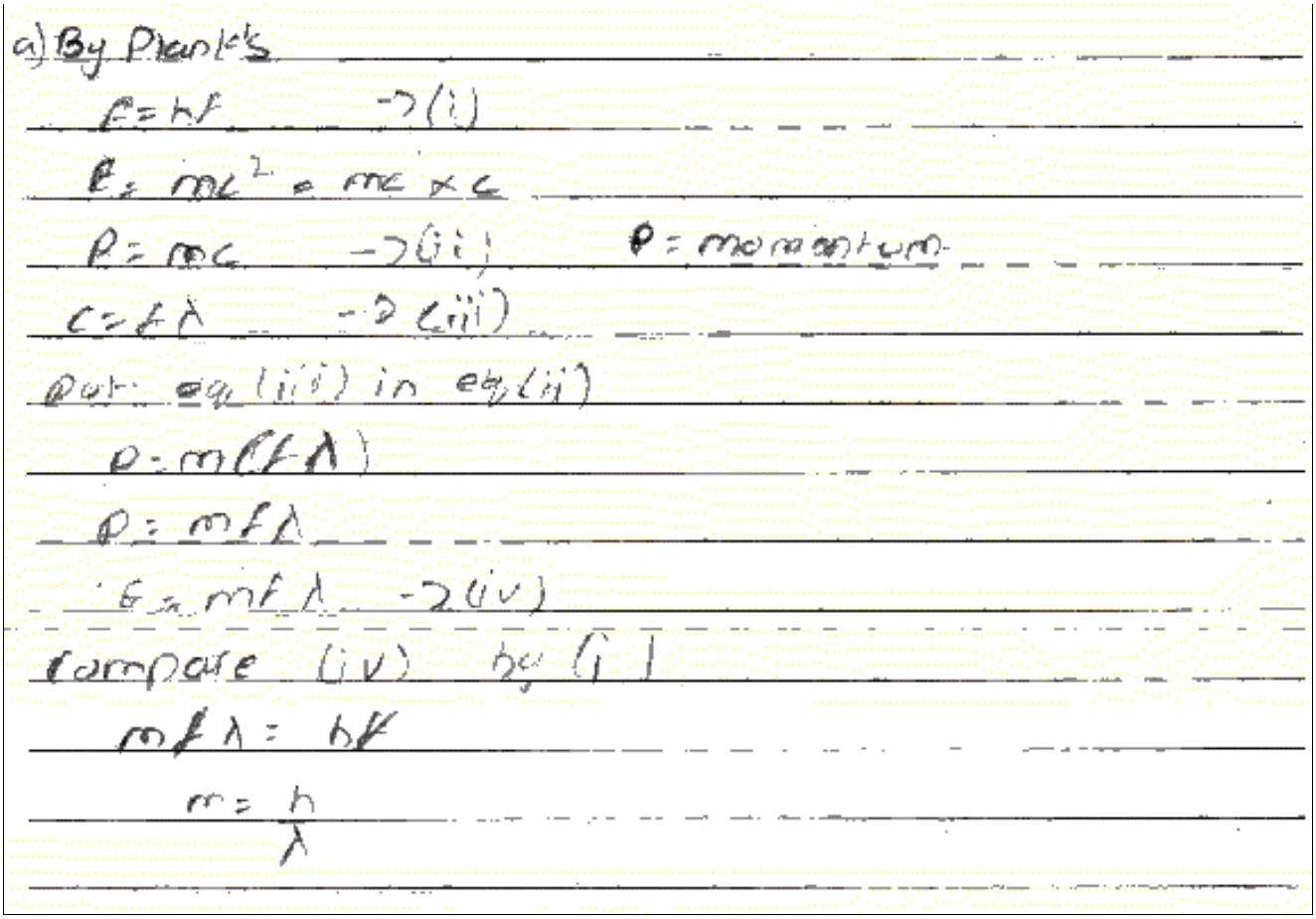
•) Different intensities of light gives ~~a~~ the same value of V_0. Flow of current increases as the intensity increase. Different lights of different frequencies at constant current gives different values for V_0.

•) A certain minimum value of frequency is needed to emit the electrons no matter how high the intensity is. It is known as threshold frequency.

•) The electrons continously emit from the surface. Intensity of light only determines ~~its~~ their number.

Weaker responses failed to provide the correct mathematical evidence with respect to the demand of the question. Some of the candidates extracted Einstein's special theory of relativity equation and tried to substitute the values in it. Whereas, few of the candidates mixed up Plank's hypothesis and Einstein's special-relativity equation $E = mc^2$ which were irrelevant to the demand of the question.

Example:



Question 17b:

Describe the characteristics of photoelectric effect in FIVE points.

Better responses correctly described the characteristics of photoelectric effect. High-scoring candidates wrote that the threshold frequency depends upon the nature of the metal for photoelectric effect/ no photoelectron is emitted when the frequency of light is below the threshold frequency. Some of the candidates wrote that the speed of photoelectrons increases with the increase in the frequency of the incident light/ the number of photoelectrons emitted is directly proportional to the intensity of the incident light and a beam of light of frequency is slightly greater than the threshold frequency may cause an immediate emission of electrons.

Example:

Photo electric effect :- When the electrons are emitted out from a metal surface, it is known as photoelectric effects.

•) The electrons are only emitted from cathode when light falls on it. The photoelectric current stops flowing as soon as the incident light is stopped.

•) The maximum energy of electrons can be calculated by reversing the connections of electrode with terminal of battery. However, at some time, the current will become zero at V_o (stopping potential).

$$\frac{1}{2}mv^2 = V_oe$$

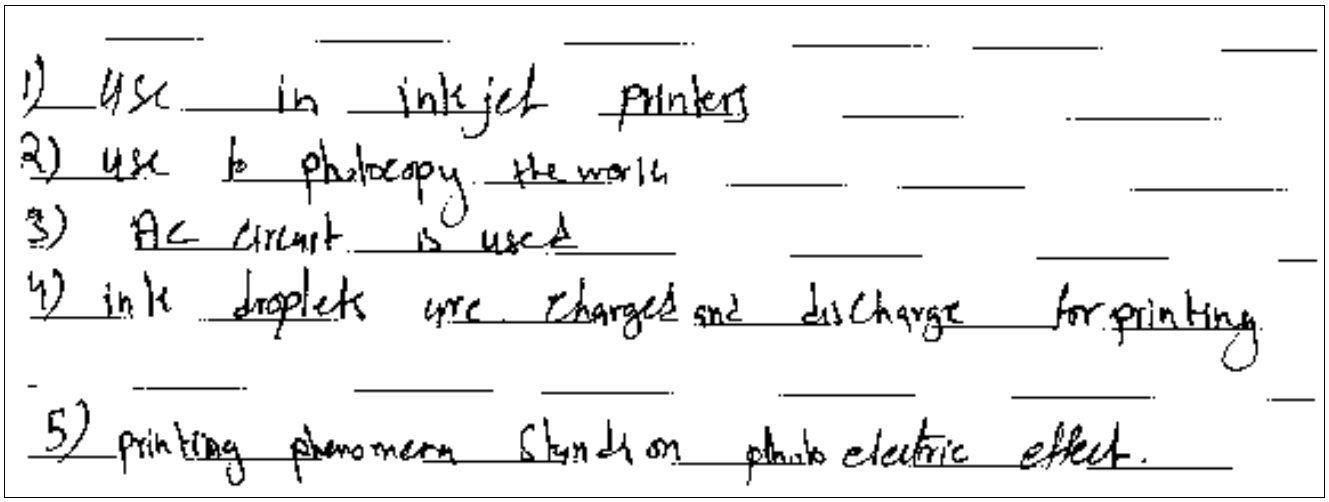
•) Different intensities of light gives ~~a~~ the same value of V_o. Flow of current increases as the intensity increase. Different lights of different frequencies at constant current gives different values for V_o.

•) A certain minimum value of frequency is needed to emit the electrons no matter how high the intensity is. It is known as threshold frequency.

•) The electrons continously emit from the surface. Intensity of light only determines ~~its~~ their number.

Weaker responses correctly wrote either one or two characteristics of photoelectric effect. Some of the low-scoring candidates repeated the same characteristics of photoelectric effect in their description. Few of the candidates mixed up results of special theory of relativity which was irrelevant to the demand of the question and shows candidates' lack of understanding and in-depth knowledge on this most important topic of modern physics.

Example:


1) Use in inkjet printers
2) use to photocopy the work
3) AC circuit is used
4) ink droplets are charged and discharge for printing
5) printing phenomena stands on photo electric effect.